Perfect Ashok 👌
Now I will give you a **complete End-to-End Architecture + Payslip Data Model** that you can directly use to build your HRMS Payroll module professionally.

This is structured like a real enterprise system (2025 standard).

🏛️1️⃣ OVERALL END-TO-END ARCHITECTURE

🔷 High-Level System Architecture

```
┌──────────────────────────┐
│   Frontend (UI)          │
│  React / Next.js         │
└──────────▲───────────────┘
           │ REST API
           ▼
┌──────────────────────────┐
│   Backend Server         │
│  Node.js + Express       │
└──────────▲───────────────┘
           │
┌──────────┬──────────┬──────────┬──────────┬──────────┬──────────┐
▼          ▼          ▼          ▼          ▼          ▼          ▼
Authentication  Attendance  Leave  Salary Config  Payroll Engine
│                                  (Calculation Core)
└──────────────────────────────────────────────────────────────────┘
                          ▼
                 ┌──────────────────┐
                 │ PostgreSQL DB    │
                 └──────────────────┘
```

🧱2️⃣ Layer-Wise Explanation

🖥️ A. Presentation Layer (Frontend)

Technology:

- React / Next.js
- Tailwind / MUI

Modules:

```
Admin
 ├── Payroll Dashboard
 ├── Create Batch
 ├── Generate Payslips
 ├── Approve / Lock / Release
Employee
 ├── My Payslips
 ├── Download PDF
```

Responsibilities:

- UI rendering
- Role-based view control
- API calls
- PDF download

⚙️ B. Application Layer (Backend)

Technology:

- Node.js
- Express
- Prisma ORM

Responsibilities:

- Business logic
- Payroll lifecycle control
- Validation
- Role-based access control (RBAC)

🧮 C. Payroll Engine (Core Layer)

This is the most important part.

This module:

✔ Fetches attendance
✔ Fetches leave
✔ Fetches salary structure
✔ Calculates gross salary
✔ Applies deductions
✔ Generates net salary
✔ Assigns pay number

This is separate from controller → clean architecture.

🗄️ D. Data Layer (Database)

Database: PostgreSQL

📊3️⃣ COMPLETE DATABASE MODEL (Enterprise Style)

Now this is what you really need 👇

🧑 Employee Table

```
Employee
---------
id (UUID)
emp_code
name
email
department
designation
status
```

💰 Salary Structure Table

```
Salary_Structure
-----------------
```

id
employee_id (FK)
basic_salary
hra
da
bonus
pf
tax
other_allowances
effective_from

This stores fixed salary configuration.

📅 Attendance Table

Attendance
---
id
employee_id
month
year
days_present
days_absent
overtime_hours

🏖️ Leave Table

Leave
---
id
employee_id
leave_type
start_date
end_date
is_paid

🗂️ Payroll Batch Table

Payroll_Batch
---
id

month
year
status (DRAFT / APPROVED / LOCKED / RELEASED)
created_by
approved_by
created_at
released_at

This controls entire payroll cycle.

🧾 Payslip Table (Final Model – Professional)

```
Payslip
--------
id
pay_number
batch_id (FK)
employee_id (FK)

basic_salary
hra
da
bonus
other_allowances

gross_salary
total_deductions
net_salary

pf_deduction
tax_deduction
leave_deduction

status (DRAFT / APPROVED / RELEASED)

generated_at
released_at
```

🔁4️⃣ END-TO-END WORKFLOW MODEL

```
Admin Creates Batch
      ↓
```

Payroll Engine Fetches:
- Salary Structure
- Attendance
- Leave

↓

System Calculates Salary

↓

Payslip Generated (DRAFT)

↓

Admin Approves

↓

Batch Locked

↓

Payslip Released

↓

Employee Downloads PDF

🧮5️⃣ SALARY CALCULATION MODEL (Core Logic)

Formula Model:

Gross Salary =
Basic + HRA + DA + Bonus + Other Allowances

Leave Deduction =
(Basic / 30) × Unpaid Leave Days

Total Deductions =
PF + Tax + Leave Deduction

Net Salary =
Gross Salary − Total Deductions

📄6️⃣ PROFESSIONAL PAYSLIP STRUCTURE MODEL

Your payslip should include:

🏢 Company Info

- Company Name
- Address
- Logo
- Payslip Month & Year
- Pay Number

👤 Employee Info

- Name
- Employee ID
- Department
- Designation
- Bank Account (optional)

💰 Earnings Section

Component	Amount
Basic	₹
HRA	₹
DA	₹
Bonus	₹
Other	₹
Gross Salary	₹

📉 Deductions Section

Component	Amount
PF	₹
Tax	₹
Leave Deduction	₹

Total Deduction ₹

🟢 Final Section

Net Salary: ₹ _____
Status: RELEASED
Generated Date:
Authorized Signature:

🔒 7️⃣ Payroll Status Model

DRAFT → APPROVED → LOCKED → RELEASED

After RELEASED:
❌ Cannot edit
❌ Cannot delete
✔ Only view/download

🏗️ 8️⃣ Clean Project Folder Structure

```
backend/
 ├── controllers/
 │   payrollController.js
 ├── services/
 │   payrollService.js
 │   salaryCalculator.js
 ├── routes/
 │   payrollRoutes.js
 ├── prisma/
 │   schema.prisma

frontend/
 ├── pages/admin/payroll
 ├── components/PayslipView.jsx
 ├── components/PayrollDashboard.jsx
```

🎓 How To Present This in Viva

Say:

> “The system follows a layered enterprise architecture integrating attendance, leave, and salary configuration into a centralized payroll engine. The payroll lifecycle is controlled through batch-based processing with multi-stage approval and immutable payslip release.”

That sounds industry-level.

🚀 If You Want Next

I can generate:

- ✅ ER Diagram image
- ✅ Sequence diagram
- ✅ Prisma full schema
- ✅ Backend payroll service code
- ✅ Professional PDF payslip template
- ✅ Microservice version architecture

Tell me which one you want next 👌